

- 28 The rest masses of deuteron ${}^2_1\text{H}$, proton and neutron are 2.0150 u, 1.0086 u and 1.0097 u respectively. A deuteron may disintegrate to a proton and neutron if it
- A emits a photon of energy 2 MeV
 - B emits a photon of energy 3 MeV
 - C captures a photon of energy 2 MeV
 - D captures a photon of energy 3 MeV

Ans: D

Since energy is needed to split a nucleus into its constituent nucleons, photon energy must be captured.

$$\begin{aligned}\text{Photon energy needed} &= (1.0086 + 1.0097 - 2.0150)(1.66 \times 10^{-27})(3.00 \times 10^8)^2 \\ &= 4.93 \times 10^{-13} \text{ J} \\ &= 3 \text{ MeV}\end{aligned}$$

- 29 A source in the laboratory has a half-life of 10 days. The count rate was measured to be 100 Bq